NullahNet

Retrofit Mumbai's nullah trash booms: time clearance to the tide, test a boom that keeps its catch, and pay on evidenced kilograms

Team: ‹TEAM NAME› · **Team ID:** ‹ID› · **Theme 3:** Rising plastic pollution in coastal cities
‹Member 1 — role› · ‹Member 2 — role› · ‹Member 3 — role› · ‹Member 4 — role› · ‹email / phone›

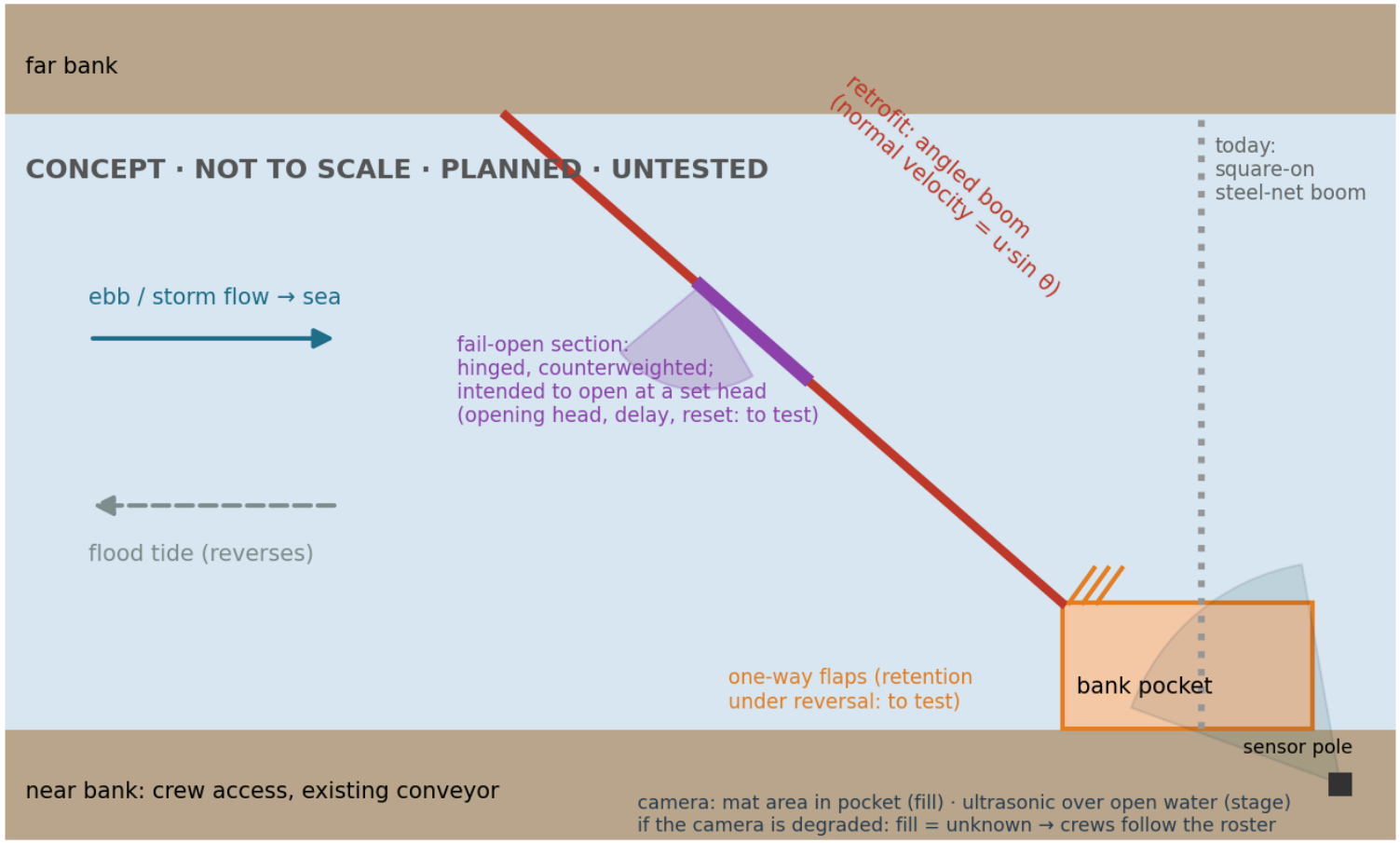
BUILT software we ran · SIMULATED digital-twin output · SYNTHETIC generated test data · SENSITIVITY assumption-driven · PLANNED not built yet

The leak, and the plastic we target

What	Value	Source
Plastic released to the sea by the metropolitan region	an estimated 5 million kg a year	The Ocean Cleanup survey, 10 Feb 2026 (an estimate)
Main pathways	"around 50 drains and nullahs"	Lokmat Times (2026), reporting the same study
What one wet spell brings back	BMC removed 2,498.66 t from seven beaches in 11 days; BMC said the rain washed refuse "from drains and natural waterways into the Arabian Sea" and "high tides then carried the waste back"	Mumbai Live, Jul 2026
What the litter is	Plastic was 75.5 % of 52,770 beach litter items (Juhu, Aksa)	ICAR-CIFE study via HT, 2023

Our source: floating single-use *flexible* packaging (carry bags, films, multilayer sachets). It has almost no resale value: waste-picker groups report it is disregarded and sometimes given to cement kilns for free (Break Free From Plastic, 2023). It is light, so storms flush it into the nullahs.

NullahNet in one picture CONCEPT · PLANNED



Every morning: the ward engineer sees each boom's fill and tomorrow's tide and rain, and gets a clearance plan. **Crews** clear in safe tide windows. **Each clearance** produces a signed record (before/after photos, weighbridge slip, GPS) that is checked against the sensors. **The retrofit** (angle, one-way pocket, fail-open section)

Every number: research/fact-check.md or eval/RESULTS.md

Mumbai already intercepts. Our gap is a hypothesis to test.

Who	What they do	Scale (source)
BMC	Steel-net booms with conveyors at nullah mouths, since July 2022 on NGT directions; cleared daily by contractors	A May 2025 report: 9 booms, expanding to 23, about ₹1 crore each, 1.5 t/day dry and 2.5–3 t/day monsoon per boom (HT; current count UNVERIFIED). Operations and maintenance: ₹14.5 crore for two years across six rivers and nullahs (FPJ, Jul 2026)
The Ocean Cleanup + Plastic Fischer	Interceptors at Malad and Trombay creeks (MCGM permits, MPCB MoU)	61 to 92 tonnes a year expected from the first systems, ≈ 1.2–1.8 % of the estimate
Plastic Fischer	TrashBooms, including one at Usha Nagar, Bhandup; kilograms verified by third-party software	3.2 million kg across 8 sites in India and Indonesia

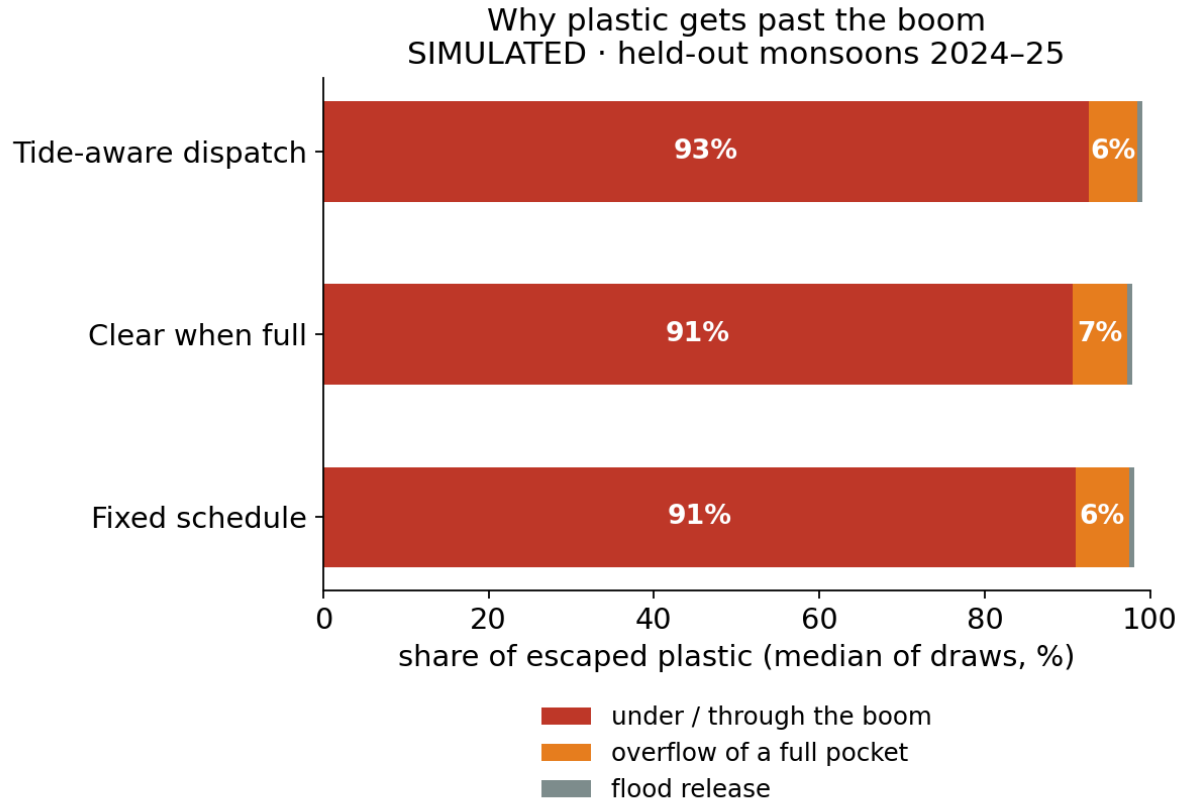
What we propose to test: we found no public evidence that BMC's booms report their fill or are cleared by tide and forecast. Our hypothesis is that adding both, plus records checked against sensors, improves this existing workflow. Context, not proof of any contractor's conduct: in the Mithi desilting case, a police SIT's FIR alleges a loss to BMC of more than ₹65.54 crore (under investigation; The Hindu, May 2025).

Why booms leak, according to our twin SIMULATED

Digital twin of four Mumbai outfalls: real rain, archived forecasts, fitted tide; assumed plastic and boom behaviour; held-out monsoons 2024–25; 40 sampled parameter settings.

- **Assumed capture losses dominate:** 92.5% of escaped plastic passes under or through the boom (central 90% scenario range 64.6%–100.0%). This includes a loss the model assumes even in still water.
- Overflow of a full pocket: 5.9% · flood release: 0.7%
- Oil-spill boom guidance (an analogy, not a plastic threshold): booms lose their catch above about 0.35 m/s across the boom, irrespective of skirt depth

So we test capture first: boom angle and pocket retention, in a tow tank, before anything else.



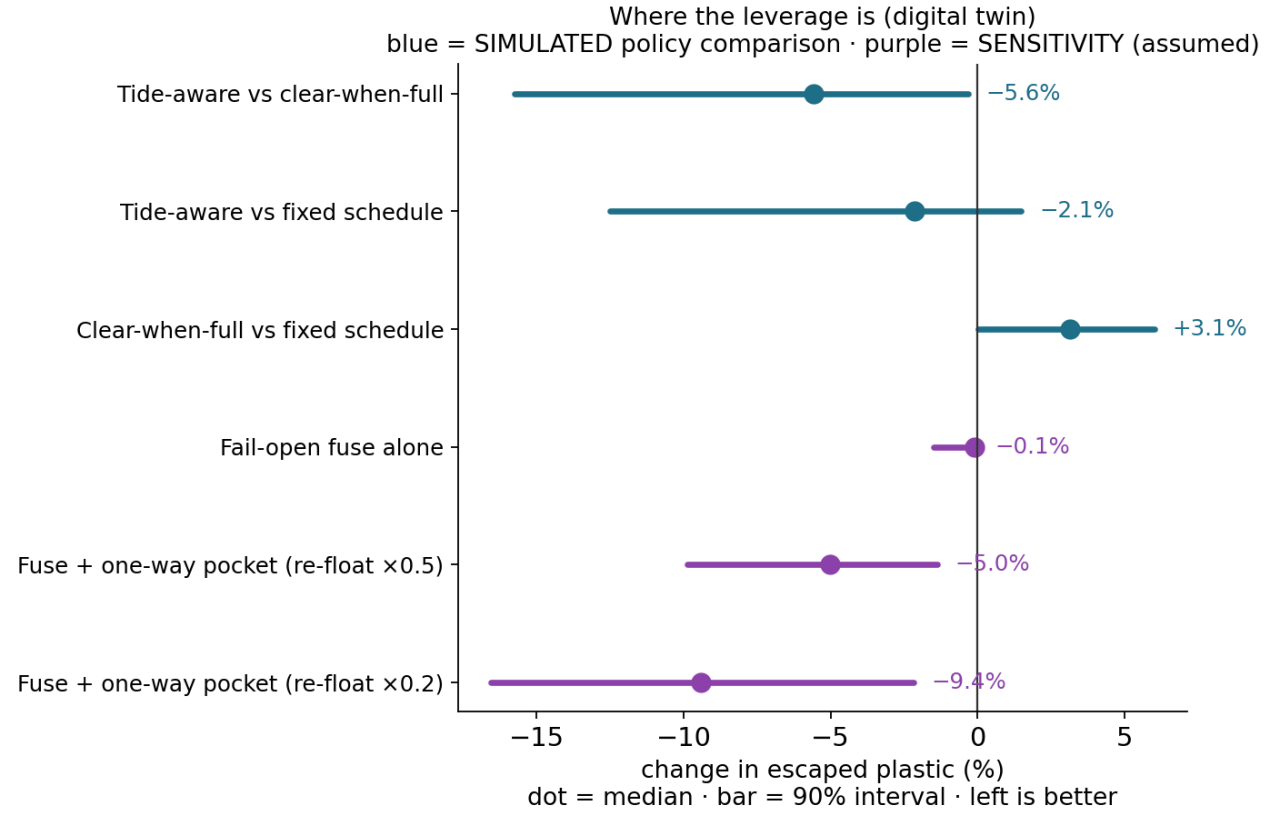
'Under the boom' includes the leak the twin assumes even in still water. Source: components/twin_sim/results/analysis.json

Evidence 1 · Timing the crews

SIMULATED

Held-out monsoons 2024–25; equal crew-hours (2 crews x 8 h/day, all runs); paired over 40 sampled parameter settings of one model.

- **The full policy** (rank booms by kilograms at risk; time crews to tide and forecast) vs **a fill-fraction-ranked fixed-shift rule:** escaped plastic **–5.6%** (central 90% scenario range –15.8% to –0.3%); lower in 39 of 40 settings
- vs **an assumed fixed roster** (not BMC's real schedule): –2.1% (range –12.5% to +1.5%), small and uncertain
- Fill-fraction rule vs assumed roster: +3.1%
- Decisions use only forecasts archived before the fact



Paired over the same Monte Carlo draws, held-out 2024–25. Source: components/twin_sim/results/analysis.json

Evidence 2 · Retrofit levers SENSITIVITY

What-if runs in the same twin (full policy; each row relative to today's boom with the same dispatch):

- **Fail-open section alone:** −0.1%. Its job is flood safety, not capture, and that safety is unproven until we measure opening head, delay and reset.
- **One-way pocket**, assuming it halves the catch lifted out by the flood tide: −5.0% (range −9.9% to −1.4%); assuming a fifth: −9.4%.
- The retention factor is an **assumption**; the tow tank must measure it. Rows are separate what-ifs: don't add them up.

Evidence 3 · Sensing and records

Detector BUILT: YOLO11n trained on public floating-litter images (TUD-GV): on 254 held-out test images, mAP50 0.857 (off-the-shelf: 0.394), precision 0.845, recall 0.784. At another site's camera (Groningen, 63 images) mAP50 falls to 0.524: it does not yet transfer, and Mumbai nullah film is untested. Public river-litter images are not Mumbai nullah film; we treat detection as an aid, not proof.

What measures fill: the camera sees the mat area in the pocket; an ultrasonic sensor over open water gives water level. If the camera is degraded (night, rain, glare), fill is reported **unknown** and crews follow the roster.

Evidence records SYNTHETIC test data: across our synthetic fraud mixture (inflated, phantom and wetted claims), 71.6% caught at 4.7% false flags. Inflated claims alone: 69.4%; small inflations only 24.4%. If debris is a quarter denser than assumed, honest false flags rise to 17.8%, so density must be calibrated per site. Tampering is caught 100% only with an external anchor; the chain alone misses tail truncation (0%). A flag triggers **review, never automatic penalty**.

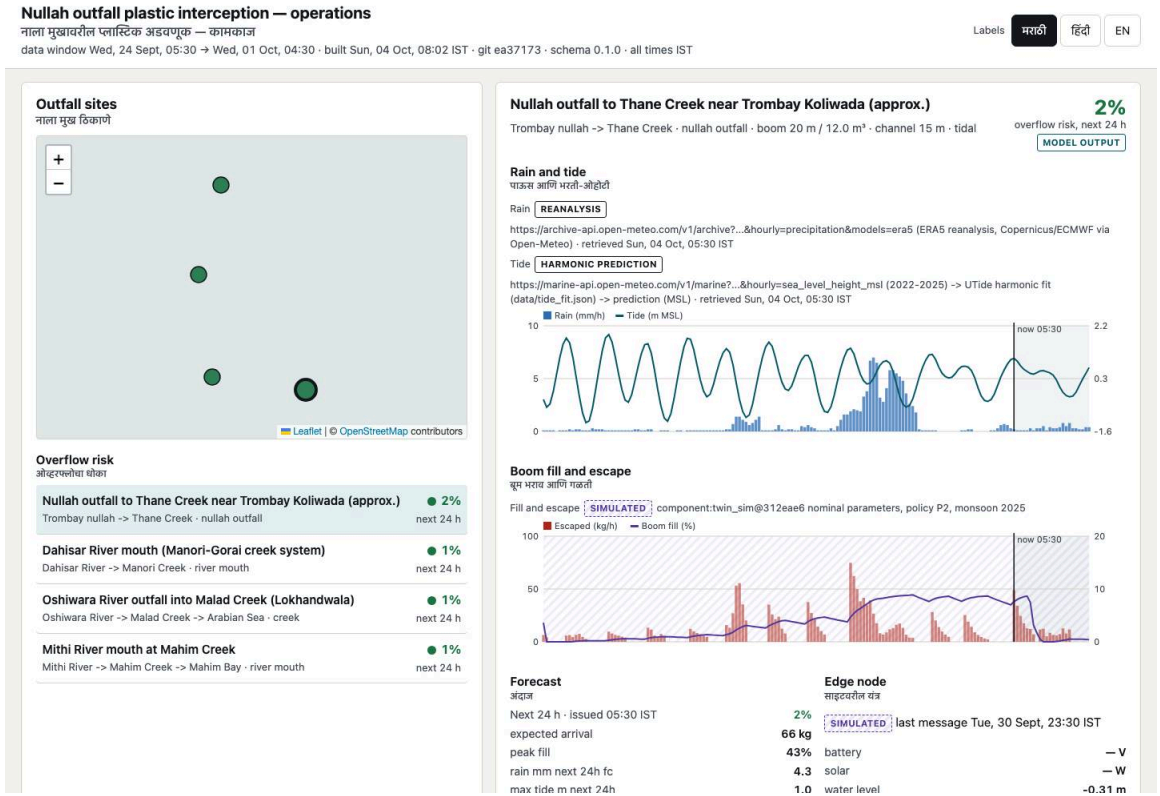
Built for Mumbai's conditions, priced from real parts

Challenge	Design answer (status)
Tides reverse the flow twice a day	Twin models each site's tide; dispatch uses tide windows (BUILT); one-way pocket to keep the catch on the flood tide (PLANNED, test T2)
Monsoon surges	Fail-open section intended to limit extra head rise within a tested flow and blockage envelope (PLANNED, test T3)
Salt, silt, hyacinth	HDPE floats, SS316 fittings, node on a pole above the high-water mark; floating traps already use HDPE and marine aluminium (Bandalong)
Monsoon power	Sized for July: 3.79 kWh/m²/day vs 7.09 in May (NASA POWER); the load is an assumption until bench-measured
Cost (listed parts only)	Sensor node ₹83,481.98, enclosure included, ≈ 0.8 % of one boom's reported price. Excludes installation, anchoring, the retrofit itself, connectivity and servicing

What we will deliver, the pilot we ask for, who pays

- **By 1 Dec (committed):** a closed-loop rig, with no field permissions needed. A tow-tank boom model (straight, angled, pocket, fail-open) + the sensor node + the twin + signed records, with measured retention, head rise and fill error.
- **Field pilot (conditional on permission, safety and a site survey):** one existing open-tidal BMC boom first, e.g. the ML Complex nullah, Dahisar, which a 2025 report says catches the most (highest volume). It is unsurveyed and uncontacted, and its gate status is UNVERIFIED. Gated outfalls come later.
- **Who pays:** the existing boom maintenance budget (₹14.5 crore for two years); EPR is upside (Category II covers "carry bags, plastic sachet or pouches", but drain-plastic eligibility is not settled); verified-kilogram credits already sell (Plastic Fischer).
- **Hook:** BMC's proposed ₹13,000 crore flood plan adds 8 to 10 feet mesh screens and expands water-level sensors beyond seven pilot sites. Compatibility with those works is UNVERIFIED.

Backup · The dashboard BUILT · values SIMULATED



Real rain (ERA5) and tide (harmonic fit) with provenance badges; boom fill and escape hatched SIMULATED; Marathi and Hindi labels; runs offline except map tiles.

Backup · How the twin works and what it leaves out

- **Inputs (real):** hourly ERA5 rain; JMA day-ahead forecasts as archived (the value issued 24 h before); tide from a UTide fit to Open-Meteo Marine model sea level. Fit RMSE 0.08 m at the Mithi is an in-sample residual against model data (fit window through Sep 2025), not gauge validation.
- **Process (assumed):** litter builds between storms and washes off with rain; capture falls with flow speed; pockets overflow; the flood tide lifts part of the catch; crews clear.
- **Uncertainty:** the main process parameters are drawn across ranges in 40 settings; site geometry, crew rules and some structures are fixed. Policy tuning used 2019–2023 only.
- **Circularity:** escaped kilograms come from the same assumed model the policies act on. Results guide experiments; they are not field evidence.
- **Not modelled yet:** flap-gated outfalls, travel time between sites, minimum clearance duration, a perfect-foresight upper bound.

Backup · Limits we state up front

Claim we do not make	Why
Tonnes diverted per year	The twin's absolute kilograms rest on assumed load and capture curves
The pocket keeps its catch	A SENSITIVITY until the tow tank measures retention under reversal
The boom cannot raise flood levels	The fail-open section's opening head, delay, fouling and reset are untested
The camera counts all plastic	Night, rain, glare and submerged film; fill is "unknown" when the camera is degraded
Item counts are accurate	On SYNTHETIC drifting-item video, the line-crossing counter over-counts by 33% (cause not yet diagnosed); counts are not used for kilograms
Records stop fraud	Hashes make edits detectable; they cannot make inputs true; flags trigger review
Drain plastic earns EPR certificates	Not found in the rules; upside only

Backup · More Mumbai context

- An independent global model ranks Malad Creek #30 of 31,819 modelled outfalls (2,724 t/yr, modelled, not measured; Meijer et al. 2021)
- A historical BRIMSTOWAD summary lists 186 outfalls, with legacy drains built for about 25 mm/h of rain at low tide; BMC's 2026 plan says drains handle about 50 mm per hour today, aiming at 100–120 mm per hour
- The 2018 plastic ban is weakly enforced: in 2024 BMC acted against 44,448 establishments and confiscated 2,148 kg
- An older BMC estimate: 500 t/day of waste goes uncollected and "ends up in the storm water drains"

Backup · Sources

- The Ocean Cleanup, press releases, 10 Feb and 30 Apr 2026 · Lokmat Times (2026) · Plastic Bank river pages (Meijer et al. 2021)
- Hindustan Times: trash booms (May 2025), beach litter (2023) · Free Press Journal, 30 Jul 2026 · Mumbai Live (2026) · The Hindu, 7 May 2025 · Citizen Matters
- ITOPF TIP 3 (booms) · NASA POWER climatology · Bandalong / Storm Water Systems · Break Free From Plastic (2023) · Ohmsett (TEAMER)
- PWM Rules 2016 as amended (UNEP LEAP text) · Pinwheel (Plastic Fischer)
- Every number with URL, quote and who opened it: `research/fact-check.md`. Every result with its reproduce command: `eval/RESULTS.md`.